OLIVIA GUO

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EDUCATION

University of California, Berkeley | May 2027 *B.S. Electrical Engineering & Computer Science*
Relevant Coursework: Digital Design (ASIC), Computer Architecture, Analog Integrated Circuits, Microelectronics, Machine Structures, Data Structures and Algorithms, Signals & Systems (FA26), Operating Systems (FA26)
Tools: Python, Git, Jira, Agile, SystemVerilog, RISC-V, C, LTSpice, Cadence, KiCAD, Oscilloscope/Power Supply/DMM

WORK EXPERIENCE

Pfizer Pearl River, NY
Software Engineer Intern May 2026 - Aug 2026
- Owned a reagent inventory tool from discovery to MVP: interviewed scientists across 15 labs, cut an initial wishlist down to a shippable scope with the dev team, and launched a Django app now tracking 1,000+ chemicals.
- Scoped and built a metrics dashboard (Appian, SQL) for Pfizer Quality: gathered requirements from 25+ stakeholders, consolidated 25 API sources into one view, and eliminated 30+ hrs/month of manual compilation.

Spiro Robotics Castro Valley, CA
R&D Systems Engineering Intern May 2025 – Aug 2025
- Spearheaded a position tracking system for device motors after identifying inefficiency caused by inconsistent positioning: used Arduino C++ scripts for low-cost analysis revealing 50% decrease in packaging time.
- Designed Python manufacturing scripts to validate 50+ device functionalities per ISO 7376 and FDA 510(k) standards.

Berkeley Imaging Systems Laboratory Berkeley, CA
Undergraduate Manager & Researcher Apr 2024 – Present
- Led a cross-functional team of 15 undergraduates (majors: EE, CS, BioEng, Biology) to purify and analyze 100+ imaging tracers, coordinating shifts, delegating tasks by interest and strengths, and presenting findings to the lab.
- Collaborate with 3 students to bring-up the first-stage analog front-end for Magnetic Particle Imaging: simulated frequency response in LTSpice, designed in KiCAD, characterized with test bench equipment.

PROJECTS

5-Stage Pipelined RISC-V CPU with I/D Caches – SystemVerilog [2-person] Spring 2026
- Optimized central processing unit architecture around a throughput target while keeping area and power constant.
- Prioritized cache efficiency over area and power, causing a 60% cycle reduction, boosting class-defined figure of merit.
- Traded complexity increases for a 35% area reduction and 10% power decrease, shutting down registers not in use.

LEADERSHIP

UC Berkeley Electrical Engineering & Computer Sciences Berkeley, CA
Undergraduate Course Instructor - Structure & Interpretation of Computer Programs (CS61A) Sep 2024 – Present
- Author exam, homework, and project content for a 1,500-student course; teach weekly sections of 30 and run office hours, translating abstract concepts (recursion, SQL, interpreters) for students with no programming background.

Cal Club Volleyball (National Rank 3/503) Berkeley, CA
President July 2024 – Present
- Manage a self-funded program of four teams, coordinating weekly practices and tournament schedules as well as designing PR strategy for grass tournament fundraisers and merchandise sales for profit ($2000+ revenue).

Education Coalition for Asian American Representation (ECAAR) Seattle, WA
Founder and co-President May 2020 – June 2023
- Identified a gap in local affinity spaces and grew a community of 50+, leading them to curate 15+ events (100k+ audience) over 3 years, garnering 35+ community stakeholders and gaining city, state & national recognition.
- Designed and developed the 2023 ECAAR Night Market, assigning 5 managers across PR, business, activities, decoration, and education to balance the event with a $2200 budget, 20+ partners, and 300+ attendees.